

CO1-AT1-Application Based Questions

Course Code: DSA0308

Course name: Natural Language Processing

Reg No: 192324315L

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1.Resume Information Extraction System Using Regular Expressions

```
import re
resumes = [
    """Name: Ravi Kumar
    Email: ravi@gmail.com
    Mobile: 9876543210
    Skills: Python, Java, SQL
    Experience: 3 years""",

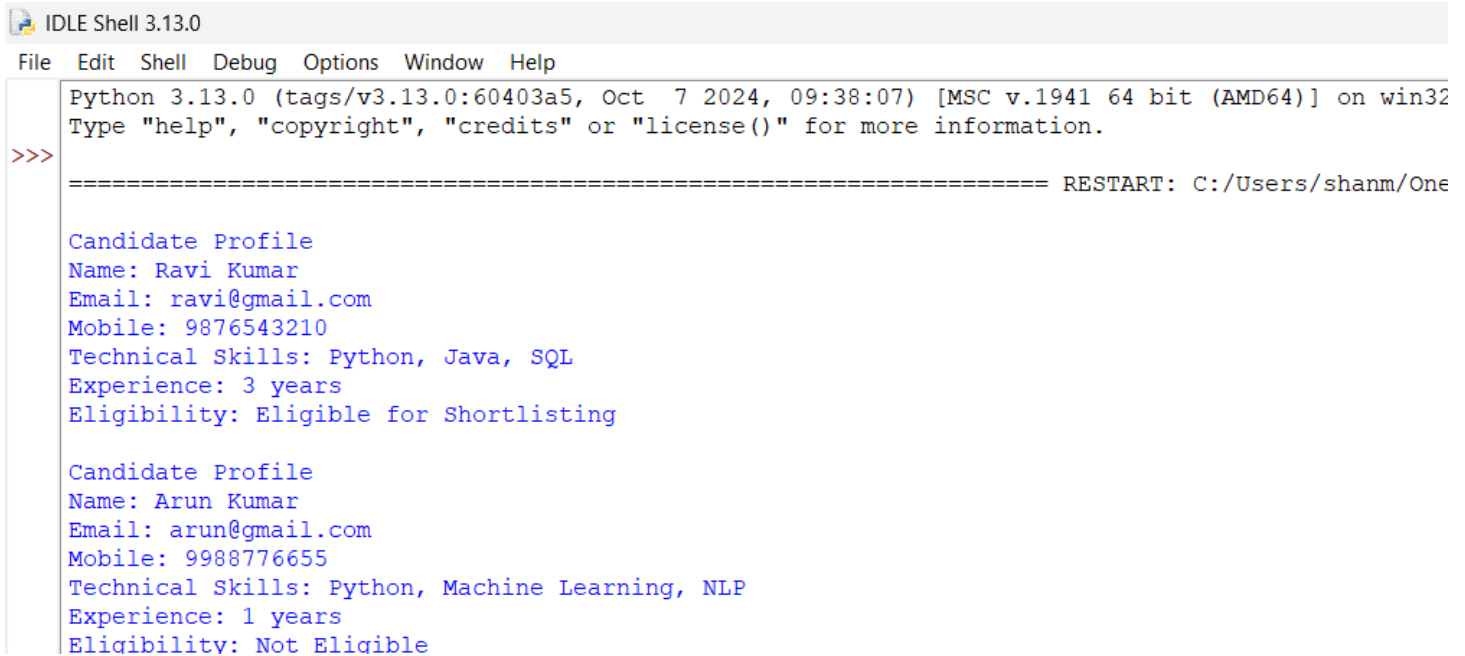
    """Name: Arun Kumar
    Email: arun@gmail.com
    Mobile: 9988776655
    Skills: Python, Machine Learning, NLP
    Experience: 1 year""",
]
for resume in resumes:
    name = re.search(r"Name:\s*(.+)", resume)
    email = re.search(r"[\w.-]+@[\w.-]+\.\w+", resume)
    mobile = re.search(r"\b[6-9]\d{9}\b", resume)
    exp = re.search(r"Experience:\s*(\d+)\s*years?", resume, re.I)
    skills = re.findall(
        r"\bPython\b\bJava\b\bSQL\b\bMachine Learning\b\bNLP\b",
        resume,
        re.I
    )
    name = name.group(1).strip() if name else "Not found"
    email = email.group() if email else "Not found"
    mobile = mobile.group() if mobile else "Not found"
    experience = int(exp.group(1)) if exp else 0
    print("\nCandidate Profile")
    print("Name:", name)
    print("Email:", email)
    print("Mobile:", mobile)
    print("Technical Skills:", ", ".join(skills) if skills else "None")
    print("Experience:", experience, "years")
```

```

    if experience >= 2 and any(s.lower() == "python" for s in skills)print("Eligibility: Eligible for
Shortlisting")
    else:
        print("Eligibility: Not Eligible")

```

Output:



```

IDLE Shell 3.13.0
File Edit Shell Debug Options Window Help
Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/shanm/One

Candidate Profile
Name: Ravi Kumar
Email: ravi@gmail.com
Mobile: 9876543210
Technical Skills: Python, Java, SQL
Experience: 3 years
Eligibility: Eligible for Shortlisting

Candidate Profile
Name: Arun Kumar
Email: arun@gmail.com
Mobile: 9988776655
Technical Skills: Python, Machine Learning, NLP
Experience: 1 years
Eligibility: Not Eligible

```

2.Product Search System Using Regular Expressions in Python

```

import re
products = [
    "Apple iPhone",
    "Samsung Galaxy",
    "Apple Watch",
    "Wireless Headphones",
    "Laptop Computer",
    "Bluetooth Speaker",
    "Apple MacBook",
    "Smart Watch"
]
keyword = input("Enter search keyword: ")
choice = input("Enter search type (exact/prefix/suffix/partial): ")
if choice == "exact":
    pattern = r"\b" + re.escape(keyword) + r"\b"
elif choice == "prefix":
    pattern = r"^" + re.escape(keyword)
elif choice == "suffix":
    pattern = re.escape(keyword) + r"$"
elif choice == "partial":
    pattern = re.escape(keyword)

```

```

else:
    print("Invalid search type")
    pattern = None
if pattern:
    matches = [
        product for product in products
        if re.search(pattern, product, re.IGNORECASE)
    ]
    print("\nMatching Products:")
    if matches:
        for product in matches:
            print(product)
    else:
        print("No products found")
    print("Search Type:", choice)
    print("Search Keyword:", keyword)
    print("Total Matching Products:", len(matches))

```

Output:

```

>>> ===== RESTART: C:/Users/shanm/OneDrive/Desktop/pl.py =====
Enter search keyword: apple
Enter search type (exact/prefix/suffix/partial): prefix

Matching Products:
Apple iPhone
Apple Watch
Apple MacBook
Search Type: prefix
Search Keyword: apple
Total Matching Products: 3
>>> |

```

3. Student Registration Validation System Using Regular Expressions in Python

```

import re
reg_no = input("Enter Register Number: ")
email = input("Enter Institutional Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester (1-8): ")
mobile = input("Enter Mobile Number: ")
v1 = bool(re.fullmatch(r"20\d{2}[A-Z]{2}\d{3}", reg_no))
v2 = bool(re.fullmatch(r"[a-zA-Z0-9._%+-]+@university\.edu", email))
v3 = bool(re.fullmatch(r"[A-Z]{2,4}\d{3}", course))
v4 = bool(re.fullmatch(r"[1-8]", semester))
v5 = bool(re.fullmatch(r"[6-9]\d{9}", mobile))
print("\nValidation Report")
print("Register Number:", "Valid" if v1 else "Invalid")

```

```
print("Institutional Email:", "Valid" if v2 else "Invalid")
print("Course Code:", "Valid" if v3 else "Invalid")
print("Semester:", "Valid" if v4 else "Invalid")
print("Mobile Number:", "Valid" if v5 else "Invalid")
if all([v1, v2, v3, v4, v5]):
    print("\nRegistration Successful")
else:
    print("\nRegistration Failed")
```

Output:

```
>>> ===== RESTART: C:/Users/shanm/OneDrive/Desktop/pl.py =====
Enter Register Number: 192324315L
Enter Institutional Email: 192324315l.simats@saveetha.com
Enter Course Code: Natural language processing
Enter Semester (1-8): 7
Enter Mobile Number: 9014290456

Validation Report
Register Number: Invalid
Institutional Email: Invalid
Course Code: Invalid
Semester: Valid
Mobile Number: Valid

Registration Failed
>>>
```